

Cross-Environment Policy Distillation: A Teacher-Student Framework for Robust Robotic Locomotion

Implementation Roadmap and Team Responsibilities

Project Objective

The objective of this project is to investigate whether knowledge learned by specialized reinforcement learning agents can be transferred into a single generalized locomotion policy.

We will train expert teacher agents on different terrains using MuJoCo Ant-v5 and then transfer their knowledge into a student policy using Policy Distillation.

The final goal is to determine whether a distilled student can outperform a student trained entirely from scratch while maintaining strong performance across multiple environments.

Project Architecture Overview

The project consists of two major branches that can run largely in parallel.

BRANCH A (Teacher Branch)

Environment Development

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Baseline Algorithm Comparison

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Teacher Training

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Teacher Models

BRANCH B (Student Branch)

Evaluation Framework

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Student From Scratch

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Distillation Framework

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Final Evaluation

The two branches merge when the teacher models become available for policy distillation.

PHASE 1 — Environment Setup and Simulation Understanding

Goal

Establish a working MuJoCo simulation environment and understand the Ant robot's observation and action spaces.

Tasks

- Install MuJoCo
- Install Gymnasium
- Install Stable-Baselines3
- Launch Ant-v5
- Run random actions
- Inspect observations
- Inspect actions
- Visualize robot movement

Deliverables

- Working Ant-v5 environment
- Environment test script
- Random policy baseline

Dependency

None

Owner

Completed by both team members

Status

Completed

PHASE 2 — PPO Baseline Training and Evaluation Framework

Goal

Create a scientifically valid reinforcement learning evaluation system.

Tasks

PPO Training

Train PPO on standard Ant-v5.

Save checkpoints periodically.

Example:

- 10k steps

- 25k steps
- 50k steps
- 100k steps
- 150k steps
- 200k steps

Evaluation Framework

Implement:

- Separate evaluation environment
- Multiple evaluation episodes
- Multiple random seeds
- Statistical reporting

Metrics

For every policy:

- Average Return
- Episode Length
- Forward Distance
- Average Velocity
- Fall Rate

Output

Create scripts that automatically generate:

- Evaluation tables
- CSV files
- Performance plots

Why This Phase Exists

Professor feedback emphasized that reinforcement learning performance is highly variable.

The best model must be selected through evaluation rather than simply choosing the latest checkpoint.

Deliverables

- PPO checkpoints
- Evaluation scripts

- Metrics collection framework
- Performance plots

Dependency

Phase 1

Owner

Person 1

Status

Partially Completed

PHASE 3 — PPO vs SAC Baseline Comparison

Goal

Determine which algorithm produces the strongest and most stable teacher policy.

Tasks

Train:

PPO Baseline

Environment:

Normal Terrain

SAC Baseline

Environment:

Normal Terrain

Evaluate both algorithms using the Phase 2 evaluation framework.

Metrics

- Average Return
- Episode Length
- Forward Distance
- Velocity
- Fall Rate
- Training Stability

Why This Phase Exists

The professor specifically requested a stronger baseline than random behavior.

This comparison strengthens the scientific validity of the project and helps justify the final teacher selection.

Deliverables

- PPO Results
- SAC Results
- Baseline Comparison Report
- Selected Teacher Algorithm

Dependency

Phase 2

Owner

Person 2

Status

Not Started

PHASE 4 — Multi-Terrain Environment Development

Goal

Create environments that introduce adaptation challenges.

Terrain A

Normal Terrain

Standard Ant-v5.

Terrain B

Slippery Terrain

Modify friction parameters to simulate icy conditions.

Tasks

- Environment wrapper creation
- Friction modification
- Environment validation
- Compatibility testing

Why We Focus on Slippery Terrain

The professor specifically referenced transfer from normal terrain to slippery or icy terrain.

This keeps the project focused and achievable within a quarter.

Deliverables

- Normal Environment
- Slippery Environment
- Terrain Documentation

Dependency

Phase 1

Owner

Person 2

Status

Not Started

PHASE 5 — Specialized Teacher Training

Goal

Train expert policies specialized for individual terrains.

Teacher A

Terrain:

Normal

Teacher B

Terrain:

Slippery

Training Algorithm

Use the best algorithm selected in Phase 3.

Possible outcomes:

- PPO Teachers
or
- SAC Teachers

Tasks

Train both teachers independently.

Evaluate both teachers using:

- Average Return
- Episode Length
- Velocity
- Distance
- Fall Rate

Deliverables

- Teacher A Model
- Teacher B Model
- Teacher Evaluation Report

Dependency

Requires:

- Phase 3 completed
- Phase 4 completed

Owner

Person 2

Status

Not Started

PHASE 6 — Student From Scratch

Goal

Create a baseline student without using teacher knowledge.

Tasks

Train a student on mixed terrain:

- Normal Terrain
- Slippery Terrain

The student learns entirely through reinforcement learning.

No distillation.

No teacher guidance.

Why This Phase Exists

The professor explicitly requested comparison against a student trained from scratch.

This becomes the most important baseline for evaluating policy distillation.

Deliverables

- Student From Scratch Model
- Student Evaluation Report

Dependency

Requires:

- Phase 2 completed
- Phase 4 completed

Does NOT require teacher models.

Owner

Person 1

Status

Not Started

PHASE 7 — Policy Distillation Framework

Goal

Build the system that transfers teacher knowledge into a student policy.

Tasks

Implement:

Teacher Policy

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Action Distribution Extraction

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KL-Divergence Loss

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Student Policy Update

Create the complete distillation pipeline.

Initially test using mock models if necessary.

The final training will use real teacher models from Phase 5.

Deliverables

- Distillation Code
- KL-Divergence Implementation
- Distillation Training Pipeline

Dependency

Implementation can begin after Phase 2.

Final training requires:

- Phase 5 completed

Owner

Person 1

Status

Not Started

PHASE 8 — Distilled Student Training

Goal

Train the final student using teacher knowledge.

Inputs

Teacher A

Teacher B

Distillation Framework

Tasks

Run policy distillation training.

Generate:

Distilled Student Model

Evaluate performance throughout training.

Deliverables

- Distilled Student Model
- Distillation Results

Dependency

Requires:

- Phase 5 completed
- Phase 7 completed

Owner

Person 1

Status

Not Started

PHASE 9 — Final Evaluation and Analysis

Goal

Determine whether policy distillation improves transfer learning and generalization.

Models To Compare

1. Random Policy
2. PPO Baseline
3. SAC Baseline
4. Teacher A
5. Teacher B
6. Student From Scratch
7. Distilled Student

Evaluation Metrics

Average Return

Overall performance.

Episode Length

Survival duration.

Forward Distance

Distance traveled.

Average Velocity

Movement efficiency.

Fall Rate

Failure percentage.

Transfer Drop

Performance reduction when moving from normal terrain to slippery terrain.

Sample Efficiency

Training required to reach target performance.

Key Research Question

Can a student distilled from expert teachers outperform a student trained entirely from scratch?

Deliverables

- Evaluation Tables
- Statistical Analysis
- Graphs
- Final Results Section

Dependency

Requires:

- Phase 5 completed
- Phase 6 completed
- Phase 8 completed

Owner

Both Team Members

Status

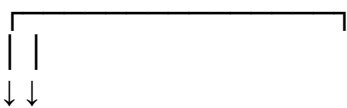
Not Started

Dependency Diagram

Phase 1



Phase 2



Phase 3 Phase 6



Phase 4



Phase 5



Phase 7

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Phase 8

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Phase 9

Parallel Work Plan

Person 1

Primary Responsibility

- Phase 2
- Phase 6
- Phase 7
- Phase 8
- Phase 9

Focus Areas

- Evaluation Framework
 - Student Training
 - Policy Distillation
 - Final Analysis
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Person 2

Primary Responsibility

- Phase 3
- Phase 4
- Phase 5

Focus Areas

- SAC Baseline
- Environment Development

- Teacher Training
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Integration Points

Integration Point 1

Phase 4 completed

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Person 1 receives terrain environments

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Student training begins

Integration Point 2

Phase 5 completed

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Teacher models become available

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Distillation training begins

Integration Point 3

Phase 6 and Phase 8 completed

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Final evaluation begins

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Project report and presentation preparation

Minimum Success Criteria

The project is considered successful if:

1. Teacher A performs strongly on normal terrain.
2. Teacher B performs strongly on slippery terrain.
3. Student From Scratch successfully learns both terrains.
4. Distilled Student successfully learns from teacher policies.
5. Distilled Student demonstrates reduced transfer drop and improved generalization compared to Student From Scratch.
6. Results are supported by statistically meaningful evaluation across multiple runs.